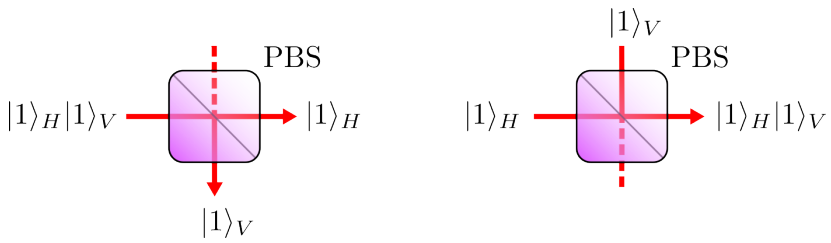

Figure 3.14: Polarising beam splitter transmitting horisontally polarised and reflecting vertically polarised light.

In practice, of course, the limit of infinite amplitude in the auxilliary mode cannot be implemented exactly. However, very good approximations to exact displacements can be realised with finite amplitudes and high-transmittance beam splitters.

Polarising beam splitters

While phase shifts and beam splitters are formally sufficient to realise any passive, linear-optical unitary, in practice of course many other optical components are needed to steer and manipulate the light in real setups, such as lenses, mirrors, fibres, and waveguides. One important type of device, not to be confused the regular beam splitter, is the *polarising beam splitter*. Polarising beam splitters allow light of one polarisation to pass through while reflecting light with orthogonal polarisation, as illustrated in Fig. 3.14. They thus allow separating light of different polarisation propagating in a shared spatial mode, or the reverse, combining light of orthogonal polarisation into one path.

Wave plates – transforming polarisation

In addition to separating polarisations, the ability to transform between different linear or circular polarisations is also crucial. This can be achived by passing the light through so-called wave plates. Wave plates are made of birefringent material that have different indices of refraction for linearly polarised light along different directions in the crystal. The effect is to shift the phase between the linear polarisation components of light passing through. By carefully choosing the length and orientation of the crystal, the phase shift can be controlled.

Half-wave plates shift the phase between linear components by π. This can be used to rotate the polarisation of linearly polarised light by a desired angle. When the angle between horisontal polarisation and the optical axis of the crystal is θ the polarisation will be rotated by 2θ. In particular, choosing $\theta = \frac{\pi}{8}$ will rotate horisontal/vertical polarisation to diagonal/anti-diagonal. It will also invert the handedness of circularly polarised light, changing left- to right-handed and vice versa.

Quarter-wave plates shift the phase between linear components by $\frac{\pi}{2}$. They are used to convert between linear and circular polarisation. When oriented with the optical axis at $\frac{\pi}{8}$ to horisontal polarisation of the incident light, they

will transform diagonal linear to right-circular and anti-diagonal linear to left-circular polarisation.

3.2.3 Encoding information in light

Light has many degrees of freedom that can be exploited to encode and manipulate quantum information. Within a single mode, all the possible states of the quantum harmonic oscillator are in principle available, and photons can be put into superposition states across different modes defined e.g. by polarisation, spatial path, frequency, time etc. Some unitary transformations and measurements will be easier or harder to realise depending on which degree of freedom is used.

Polarisation qubits

To realise a qubit, in principle any two-dimensional subspace can be used. Perhaps the most intuitive property and one that is often employed in practice is the polarisation of a single photon. Consider two modes, H and V, with identical characteristics except one is horisontally and the other vertically polarised. The two states

$$|1\rangle_H = \hat{a}_h|vac\rangle, \tag{3.148}$$
$$|1\rangle_V = \hat{a}_V|vac\rangle, \tag{3.149}$$

both contain a single photon and span a two-dimensional space. Since they have the same energy, transformations between them can be implemented using only passive, linear optics. In fact, a great feature of polarisation qubits is that any single-qubit unitary can be readily implemented with standard optical components. Measurements in any basis can then also be realised by combining this with polarising beam splitters and single-photon detectors.

For polarisation qubits, it is customary and convenient to introduce the simplified notation

$$|H\rangle = |1\rangle_H, \tag{3.150}$$
$$|V\rangle = |1\rangle_V, \tag{3.151}$$

and to define

$$|+\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle), \tag{3.152}$$
$$|-\rangle = \frac{1}{\sqrt{2}}(|H\rangle - |V\rangle), \tag{3.153}$$
$$\tag{3.154}$$

corresponding to states of diagonal and anti-diagonal polarisation, respectively, and

$$|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle), \tag{3.155}$$
$$|L\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle), \tag{3.156}$$